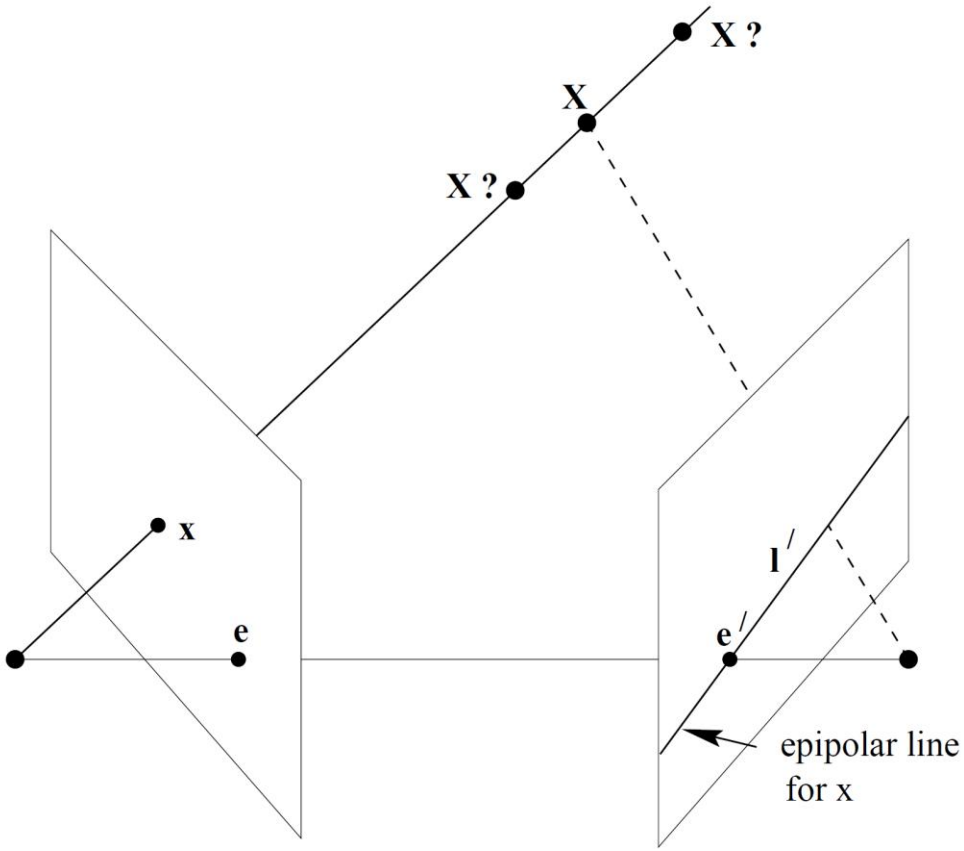
Epipolar Geometry and Stereo

CS 4391 Introduction Computer Vision

Professor Yu Xiang

The University of Texas at Dallas

Recall Fundamental Matrix



- Epipolar line $\mathbf{l}' = F\mathbf{x}$

$$\mathbf{l} = F^T\mathbf{x}'$$

- Fundamental matrix

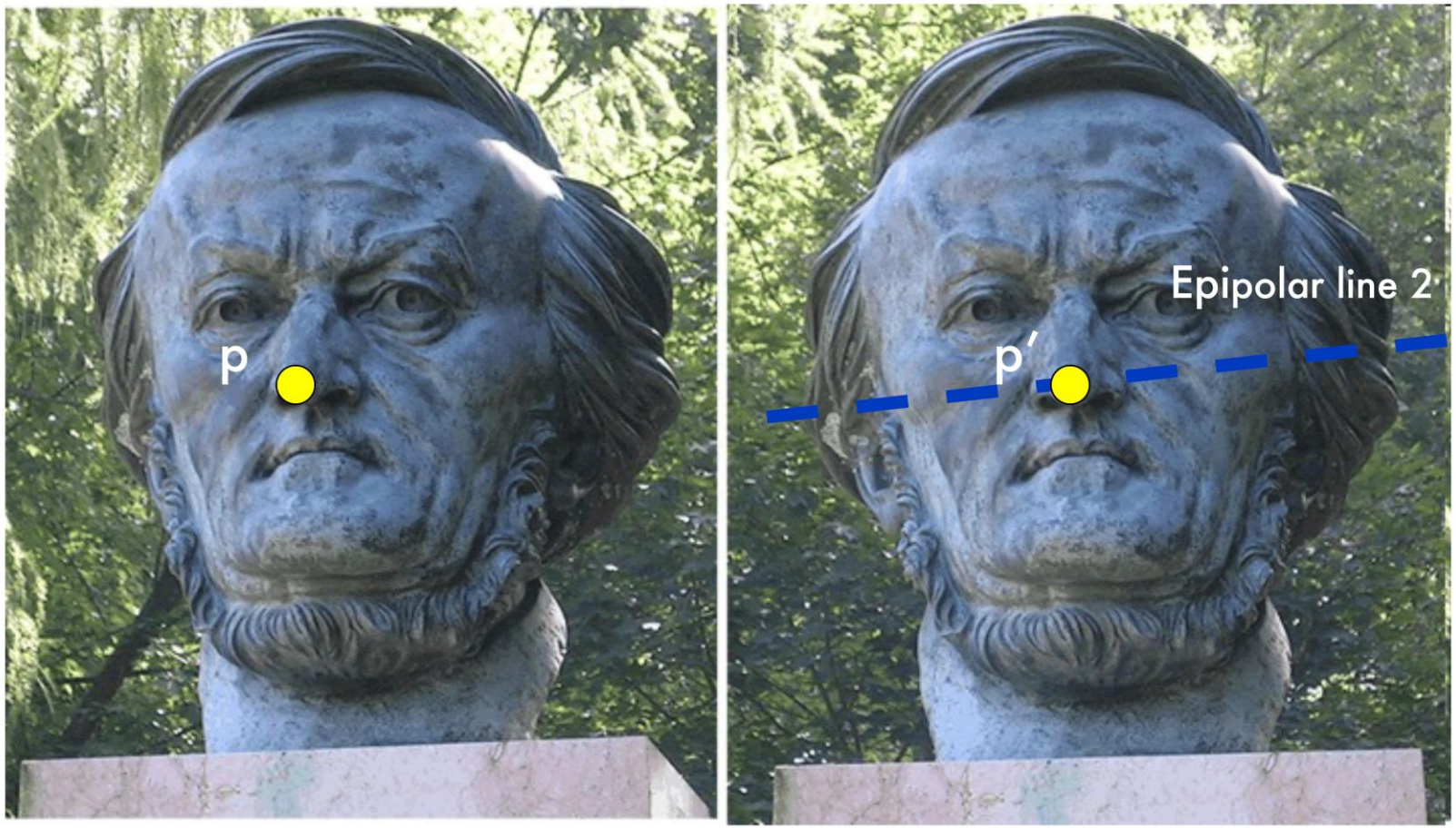
$$F = [\mathbf{e}']_\times P' P^+$$

3x3

Epipole $\mathbf{e}' = (P'C)$

$$P^+ = P^T(PP^T)^{-1} \qquad F = K'^{-T}[\mathbf{t}]RK^{-1}$$

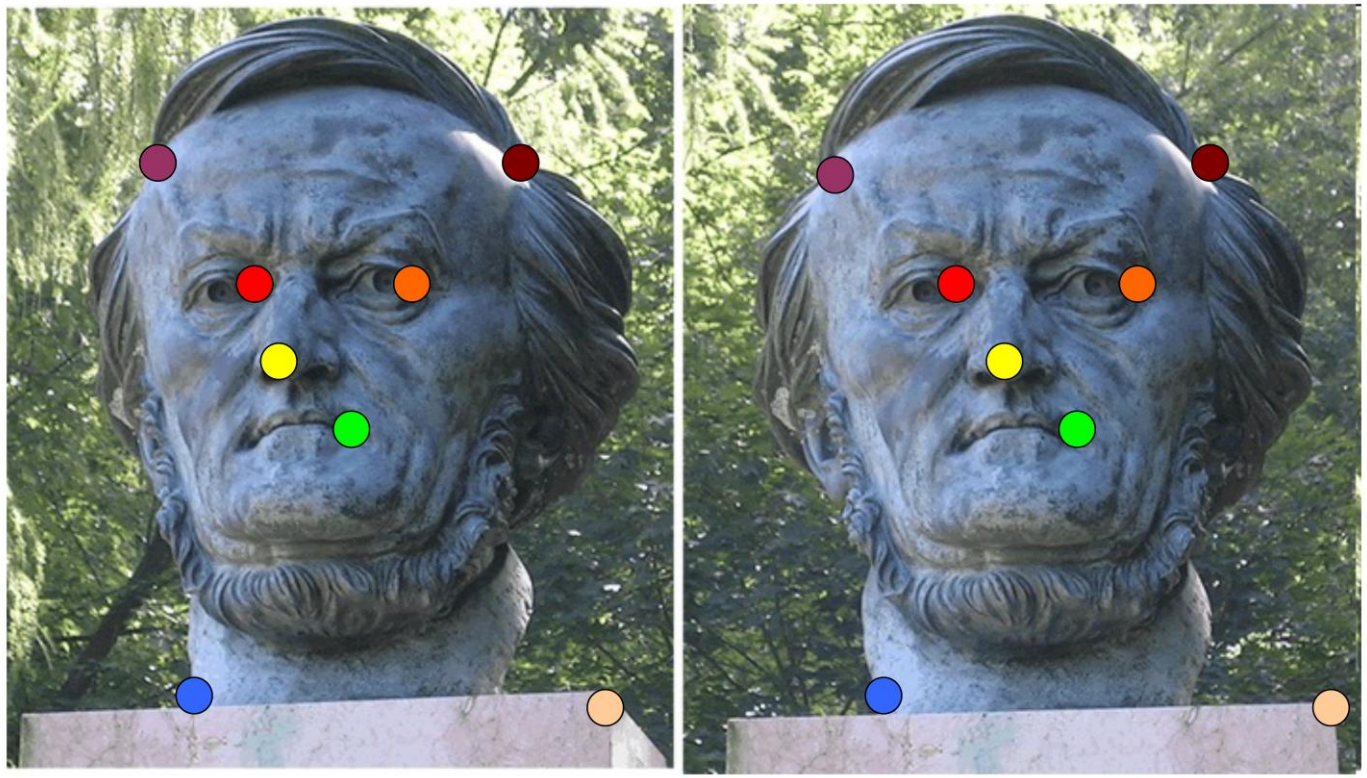
Why the Fundamental Matrix is Useful?



$$\mathbf{l}' = F\mathbf{p}$$

Estimating the Fundamental Matrix

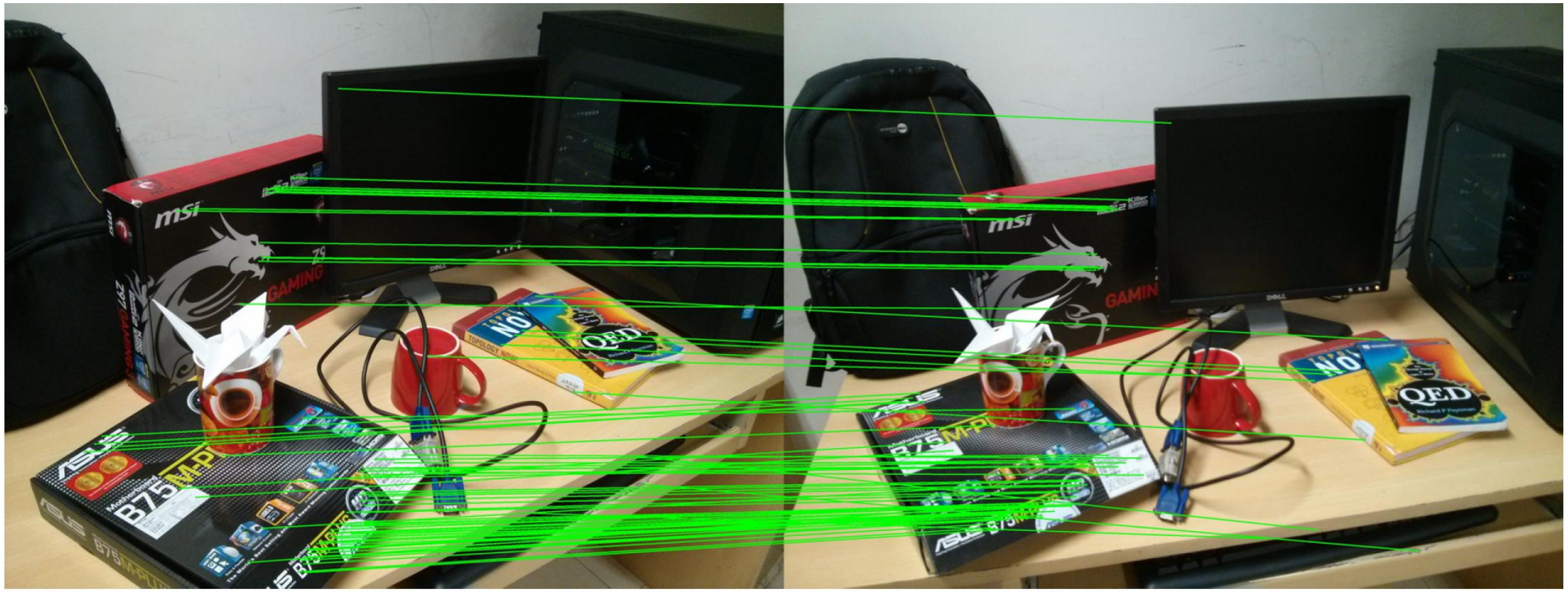
- The 8-point algorithm



$$\mathbf{l}' = F\mathbf{x}$$

$$\mathbf{x}'^{\mathsf{T}}\mathbf{F}\mathbf{x} = 0$$

Feature Detection and Matching



Geometry-aware Feature Matching for Structure from Motion Applications. Shah et al, WACV'15

Applications: stereo matching, image stitching, 3D reconstruction, camera pose estimation, object recognition

Estimating the Fundamental Matrix

$$\mathbf{x}'^{\mathsf{T}}\mathbf{F}\mathbf{x} = 0 \qquad \mathbf{x} = (x, y, 1)^{\mathsf{T}} \qquad \mathbf{x}' = (x', y', 1)^{\mathsf{T}}$$

$$x'xf_{11} + x'yf_{12} + x'f_{13} + y'xf_{21} + y'yf_{22} + y'f_{23} + xf_{31} + yf_{32} + f_{33} = 0$$

$$(x'x, x'y, x', y'x, y'y, y', x, y, 1)\,\mathbf{f} = 0$$

n correspondences

$$\mathbf{A}\mathbf{f} = \begin{bmatrix} x'_1x_1 & x'_1y_1 & x'_1 & y'_1x_1 & y'_1y_1 & y'_1 & x_1 & y_1 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \\ x'_nx_n & x'_ny_n & x'_n & y'_nx_n & y'_ny_n & y'_n & x_n & y_n & 1 \end{bmatrix} \mathbf{f} = \mathbf{0}$$

Linear System

$$A\mathbf{f} = 0$$

$$n \times 9 \quad 9 \times 1$$

- Find non-zero solutions
- If f is a solution, k×f is also a solution for $k \in \mathcal{R}$
- If the rank of A is 8, unique solution (up to scale)
- Otherwise, we can seek a solution $\|\mathbf{f}\| = 1$

$$\min \|A\mathbf{f}\|$$

Subject to $\|\mathbf{f}\| = 1$

Singular Value Decomposition

The SVD is a factorization of a $m \times n$ matrix into

$$\boldsymbol{A} = \boldsymbol{U}\,\boldsymbol{\Sigma}\,\boldsymbol{V}^{\boldsymbol{T}}$$

where $\boldsymbol{U}$ is a $m \times m$ orthogonal matrix, $\boldsymbol{V}^{\boldsymbol{T}}$ is a $n \times n$ orthogonal matrix and $\boldsymbol{\Sigma}$ is a $m \times n$ diagonal matrix.

For a square matrix ($\boldsymbol{m} = \boldsymbol{n}$):

$$\sigma_1 \geq \sigma_2 \geq \sigma_3 \ldots$$

$$\boldsymbol{A} = \begin{pmatrix} \vdots & \ldots & \vdots \\ \boldsymbol{u}_1 & \ldots & \boldsymbol{u}_n \\ \vdots & \ldots & \vdots \end{pmatrix} \begin{pmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_n \end{pmatrix} \begin{pmatrix} \ldots & \mathbf{v}_1^T & \ldots \\ \vdots & \vdots & \vdots \\ \ldots & \mathbf{v}_n^T & \ldots \end{pmatrix}$$

non-negative real numbers on the diagonal

Linear System

$$\min \|\mathbf{P}m\|$$
$$\|m\| = 1$$

Singular value decomposition (SVD)

$$P = UDV^T \qquad \|Pm\| = \|UDV^T m\| = \|DV^T m\|$$

$$\|m\| = \|V^T m\| \qquad \min \|DV^T m\| \quad \text{s.t.} \quad \|V^T m\| = 1$$

Let $y = V^T m \qquad \min \|Dy\| \quad \text{s.t.} \quad \|y\| = 1 \qquad \mathbf{y} = (0, 0, \ldots, 0, 1)^T$

$$m = V\mathbf{y}$$

m is the last column of V

Linear System

$$\underset{n \times 9}{A}\underset{9 \times 1}{\mathbf{f}} = 0$$

$$\min \|A\mathbf{f}\|$$

Subject to $\|\mathbf{f}\| = 1$

Solution: $A = UDV^T$ SVD decomposition of A

$$\underset{n \times n}{U}\ \underset{n \times 9}{D}\ \underset{9 \times 9}{V^T}$$

f is the last column of V

A5.3 in HZ

Estimating the Fundamental Matrix

- The singularity constraint $\det \mathtt{F} = 0$

$$\min \|F - F'\|$$

Subject to $\det F' = 0$

$$F = UDV^T$$

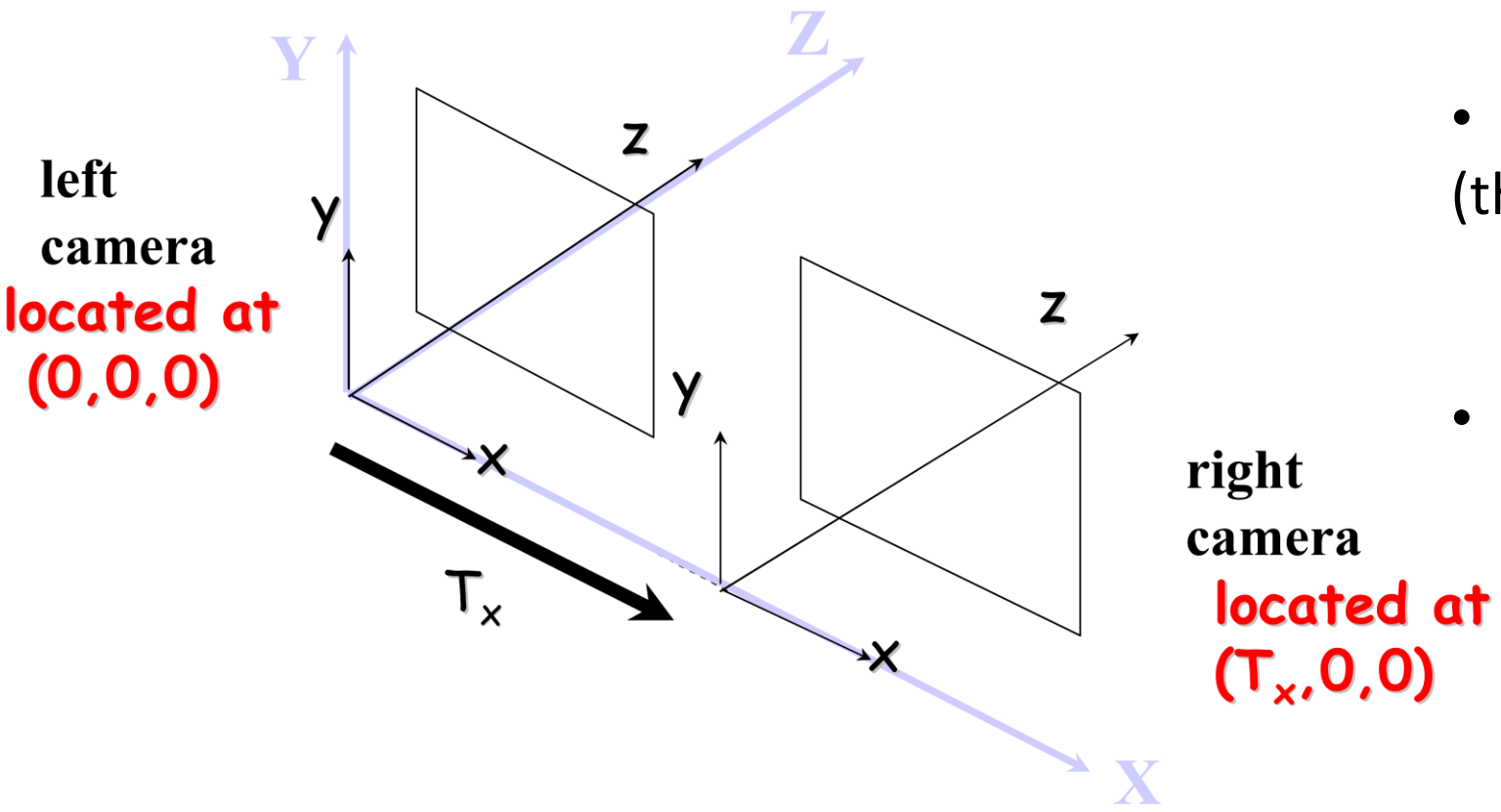
$$\mathtt{D} = \mathrm{diag}(r, s, t)$$

$$r \geq s \geq t.$$

Solution:

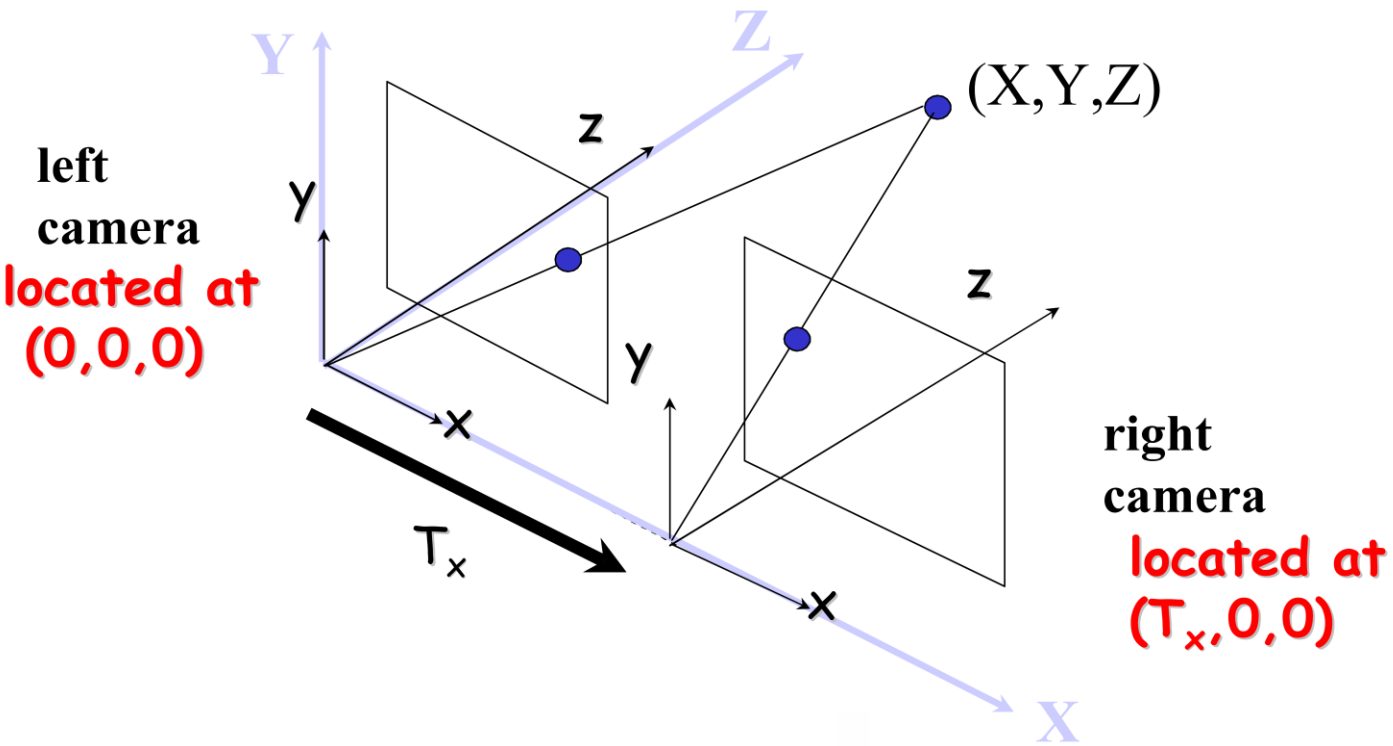
$$\mathtt{F}' = \mathtt{U}\mathrm{diag}(r, s, 0)\mathtt{V}^\mathsf{T}$$

Special Case: A Stereo System



- The right camera is shifted by T_x (the stereo baseline)
- The camera intrinsics are the same

Special Case: A Stereo System



- Left camera

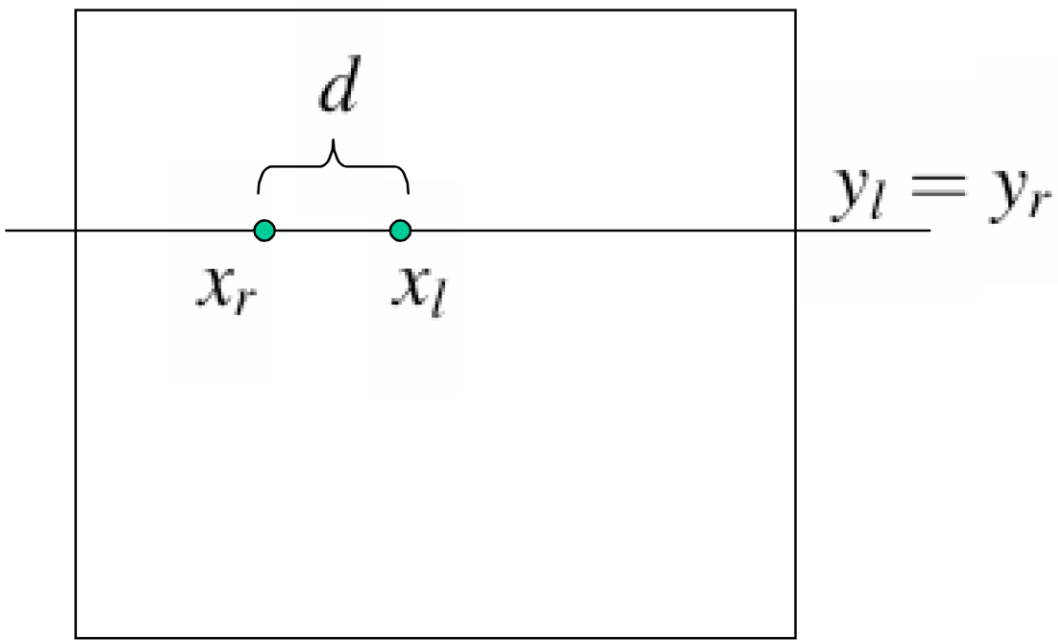
$$x_l = f\frac{X}{Z} + p_x \qquad y_l = f\frac{Y}{Z} + p_y$$

- Right camera

$$x_r = f\frac{X - T_x}{Z} + p_x$$

$$y_r = f\frac{Y}{Z} + p_y$$

Stereo Disparity



- Disparity

$$
\begin{aligned}
d &= x_l - x_r \\
&= (f\frac{X}{Z} + p_x) - (f\frac{X - T_x}{Z} + p_x) \\
&= f\frac{T_x}{Z}
\end{aligned}
$$

- Depth

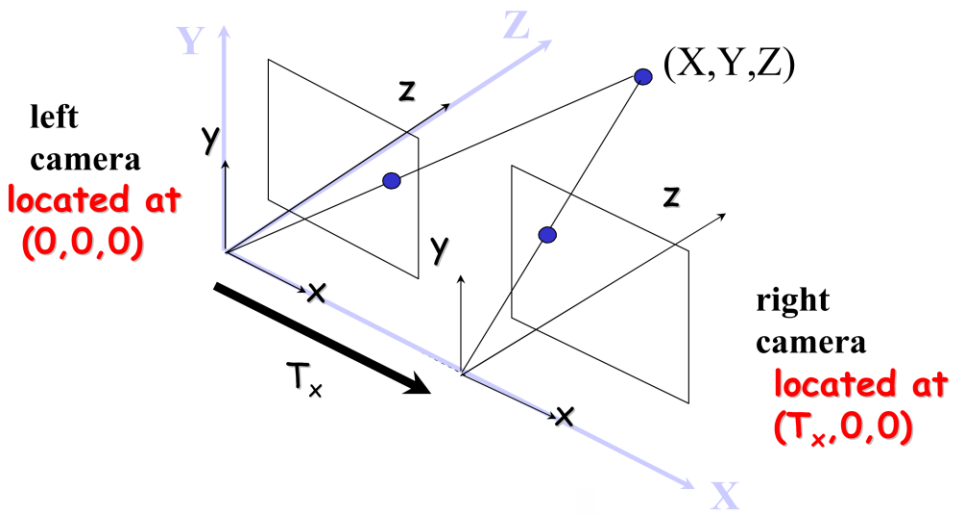
$$Z = f\frac{T_x}{d}$$

Baseline

Disparity

Recall motion parallax: near objects move faster (large disparity)

Special Case: A Stereo System



$$\mathtt{P} = \mathtt{K}[\mathtt{I} \mid \mathbf{0}] \qquad \mathtt{P}' = \mathtt{K}[\mathtt{I} \mid \mathbf{t}]$$

$$\mathtt{F} = [\mathbf{e}']_\times \mathtt{K}'\mathtt{R}\mathtt{K}^{-1} = \mathtt{K}'^{-\mathsf{T}}[\mathbf{t}]_\times \mathtt{R}\mathtt{K}^{-1} = \mathtt{K}'^{-\mathsf{T}}\mathtt{R}[\mathtt{R}^{\mathsf{T}}\mathbf{t}]_\times \mathtt{K}^{-1} = \mathtt{K}'^{-\mathsf{T}}\mathtt{R}\mathtt{K}^{\mathsf{T}}[\mathbf{e}]_\times$$

$$\mathbf{F} = [\mathbf{e}']_\times \mathtt{K}\mathtt{K}^{-1} = [\mathbf{e}']_\times$$

$$\mathbf{e}' = (P'C) \qquad \mathbf{C} = \begin{pmatrix} \mathbf{0} \\ 1 \end{pmatrix}$$

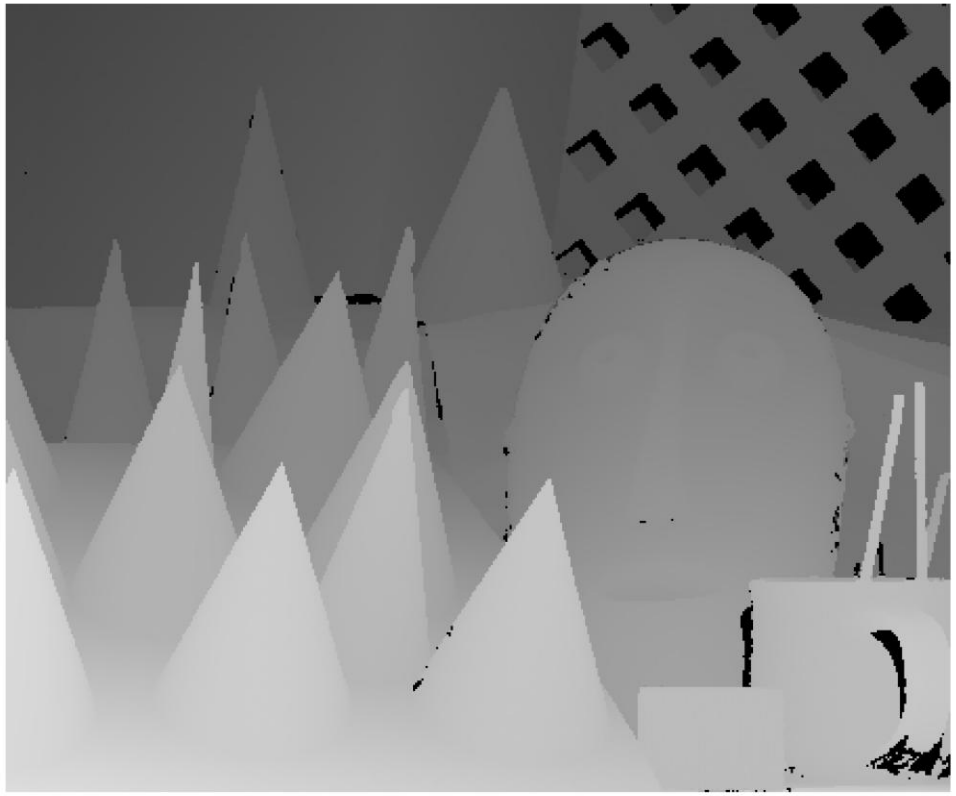
$$K = \begin{bmatrix} f_x & 0 & p_x \\ 0 & f_y & p_y \\ 0 & 0 & 1 \end{bmatrix} \qquad \mathbf{e}' = \begin{bmatrix} f_x T_x \\ 0 \\ 0 \end{bmatrix} \qquad F = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -f_x T_x \\ 0 & f_x T_x & 0 \end{bmatrix}$$

$$\mathbf{x}'^T F \mathbf{x} = 0$$

$$y = y'$$

Stereo Example

Disparity values (0-64)



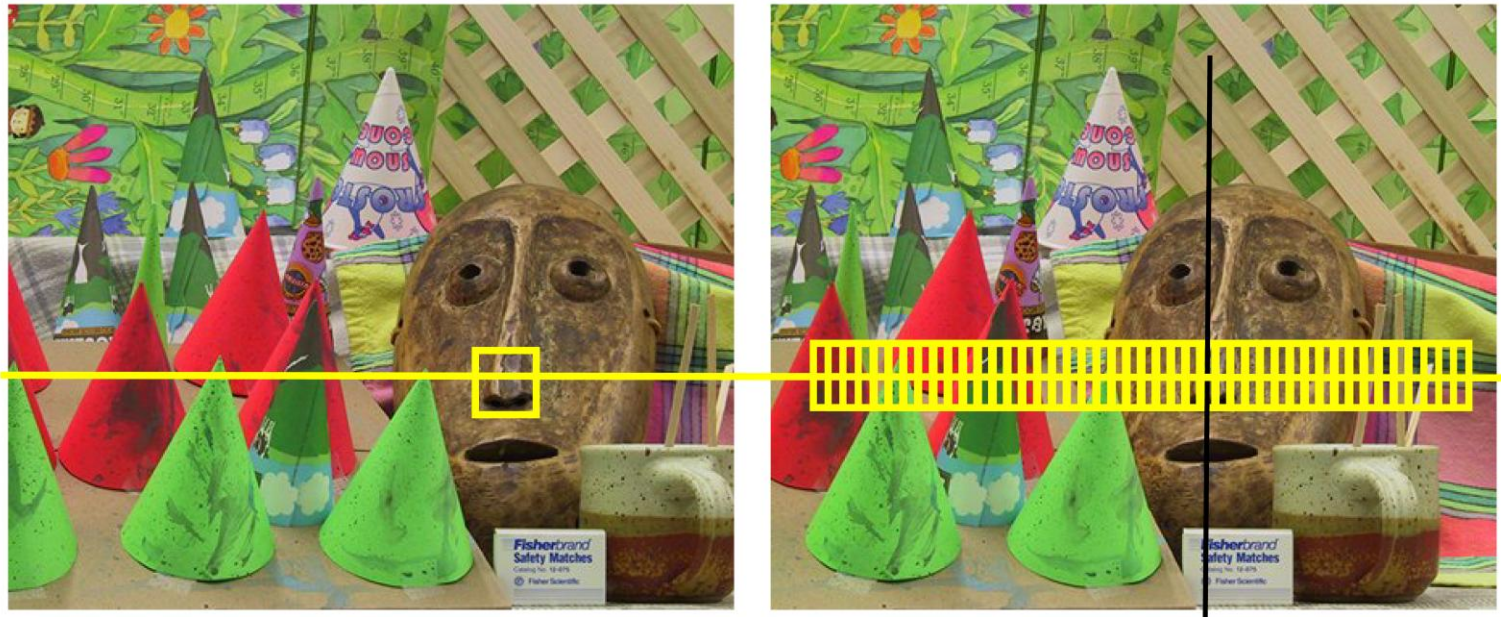
$$d = f\frac{T_x}{Z}$$

Note how disparity is larger (brighter) for closer surfaces.

Computing Disparity

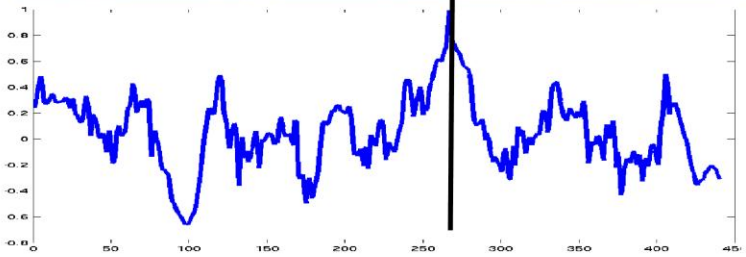
Left Image

Right Image



For a patch in left image

Compare with patches along same row in right image

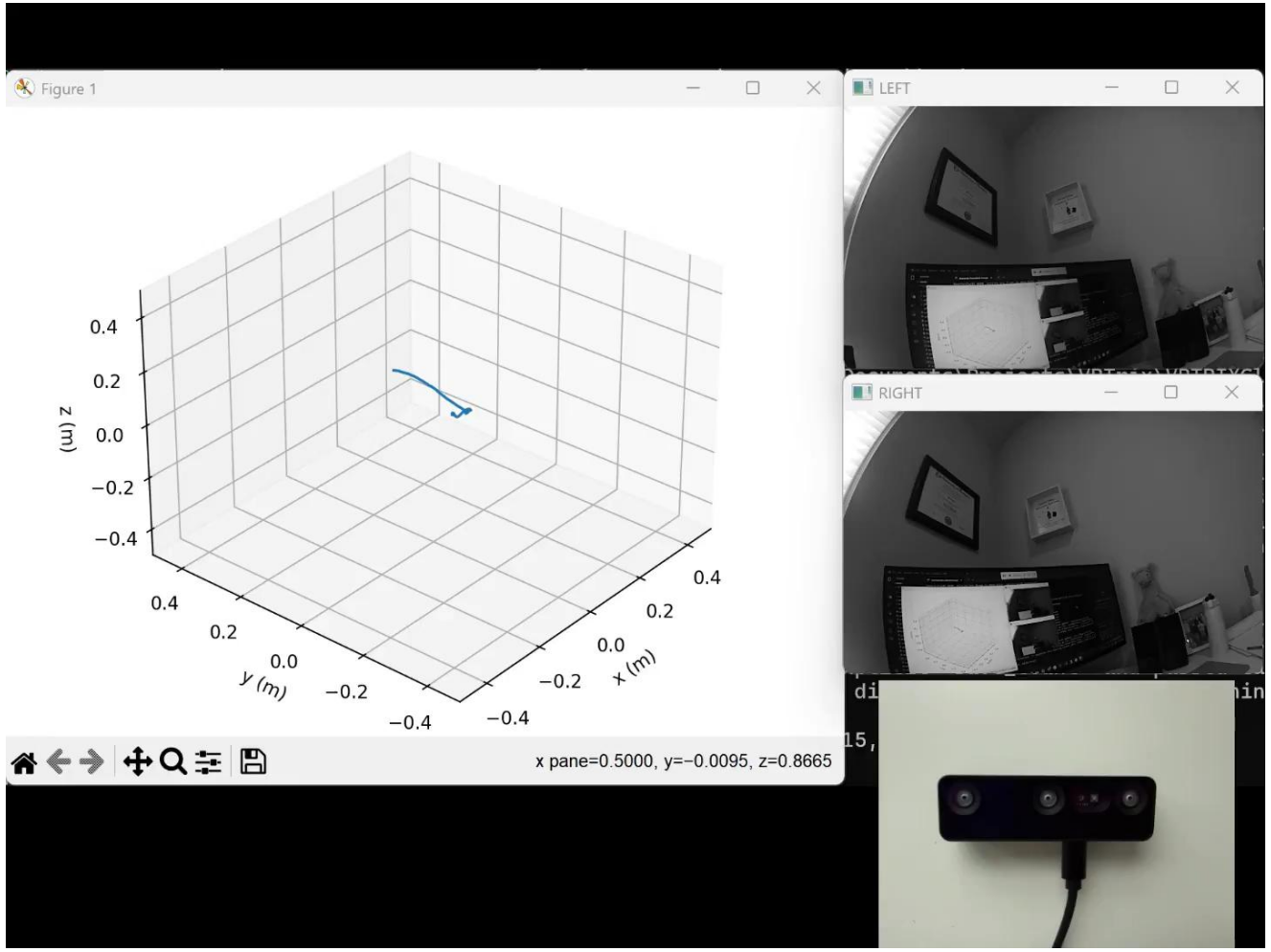


Match Score Values

- Eipipolar lines are horizontal lines in stereo
- For general cases, we can find correspondences on eipipolar lines
- Depth from disparity

$$Z = f\frac{T_x}{d}$$

Stereo Example



Further Reading

- Stanford CS231A: Computer Vision, From 3D Reconstruction to Recognition, Lecture 5 https://web.stanford.edu/class/cs231a/syllabus.html

- Multiview Geometry in Computer Vision, Richard Hartley and Andrew Zisserman, Chapter 9, Epipolar Geometry and Fundamental Matrix

- https://nvlabs.github.io/FoundationStereo/